LMIs in Control: Riccati, Dissipativity and $\mathcal{H}_\infty$

Convex Optimization, Homework 5

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English summary of my original Persian report, which is kept as a PDF in this folder.

Problems and what I did

1. **Riccati inequality as an LMI.** I wrote the Riccati inequality $A^TPA + Q - A^TPB(R + B^TPB)^{-1}B^TPA \succeq P$ as a single LMI using the Schur complement.
2. **Dissipativity of a loop with $P(s) = 1/(-3s + 1)$.** Closed-loop stability needs $k < -1$. Dissipativity with supply rate $s(w, z) = wz$ holds for $k \geq 0$ or $k \leq -1$. Together the condition is $k \leq -1$.
3. **An RLC circuit.** With states $x_1 = v_C$ and $x_2 = i$, the storage function is the stored energy, and its derivative is the supplied power minus $Ri^2 \leq 0$. So the circuit is always dissipative. For the second configuration I derived the condition as negative semidefiniteness of a 2×2 matrix in R, L and C.
4. **A second dissipativity check.** I verified the dissipation inequality for both the supply-rate and the storage-function forms.
5. **Fastest decay rate.** $\dot{V} \leq -2\beta V$ with $V = x^TPx$ leads to the generalised eigenvalue problem $\min(-\beta)$ s.t. $A^TP + PA + 2\beta P \preceq 0$, $P \succ 0$.
6. **LQR-type design as an SDP (`Q6.m`).** I multiplied by $Y = P^{-1}$, set $L = KY$ and applied the Schur complement, which gives an LMI with objective $\operatorname{tr} Y$, solved for an uncertain parameter $|\alpha| \leq 0.1$. My report gives $K \approx [1.2 \;\; 3.2 \;\; 4.1]$. Re-running the script with the current YALMIP gives $K = [1.2157 \;\; 3.2718 \;\; 4.1376]$. Because αY is bilinear, YALMIP hands the problem to `fmincon`, which stops with a feasibility warning.
7. **Passivity and bounded-real LMIs.** I wrote the LMI conditions for passivity and for an $\mathcal{H}_\infty$-norm bound.
8. **State-feedback $\mathcal{H}_\infty$ design (`Q8.m`, `Q9.m`).** I applied the bounded-real lemma with $X = P^{-1}$, $Y = KX$ and minimised γ with `lmilab`. The report gives the resulting gain for Q8, and for Q9 concludes that all eigenvalues of P are positive, so the closed loop is stable.

Status of the code: `Q6.m` runs. `Q8.m` has a missing closing parenthesis on its last line (`disp(value(Y*pinv(X)));`). `Q9.m` uses strict inequalities (`<`, `>`), which current YALMIP versions reject. Both were left as submitted.